Covering The Long Shadow

The Gap-closing Approach to Health Disparity by Social Origin in Japan

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Introduction

Health Disparity by Education on Intergenerational View

The health disparity of G2 by education (House et al. 1994; Phelan, Link, and Tehranifar 2010)

- **Parental (G1's) education** (Gale et al. 2016; Arpino, Gumà, and Julià 2018; Donnelly, Lin, and Umberson 2023; Ferraro, Schafer, and Wilkinson 2016; Luo and Waite 2005; Morton and Ferraro 2020; Williams-Farrelly and Ferraro 2023)
- **Child's (G3's) education** (Potente, Präg, and Monden 2023; Zimmer, Hermalin, and Lin 2002; Yahirun, Sheehan, and Mossakowski 2018; Jiang 2019; Wang et al. 2022; Yang, Martikainen, and Silventoinen 2016; Zimmer et al. 2007; Torssander 2013; Lee 2017)

Little literature combines two strands of research

Research Question

Figure 1: The framework in this study

How much is the health disparity of G2 by G1's education closed when All G3 are assigned to the university graduate?

- Health disparity is shaped in a multigenerational way
- Spillover effect of closing educational inequality of G3 on the health of G2

The **'gap-closing' approach** (Lundberg 2022) answers my research question

The ‘Gap-closing’ Approach

‘The expected disparity in an outcome (e.g., income) across categories of units in that sample who receive that counterfactual intervention to the treatment’ (Lundberg 2022: 2)

- **Little interference**: considering the hypothetical sample instead of the entire population
- **No counterfactual settings** in the social origin: more realistic and informative to policy

A straightforward setting to causally infer the moderating role of G3’s education on health disparity of G2 by G1’s education

Methods

Data and Sample

Social Stratification and Mobility survey in 2015 (SSM2015)

- Complete information on respondents’ work and family history retrospectively
- Useful to explicitly adjust the pre-treatment confounder

Sample

1. The respondents(G2) who have one or more children(G3) who are 25 or older
2. The respondents(G2) who are 50 or older

- Applying the listwise deletion, 2530 respondents are included

Variables: outcome and treatment

Outcome

- Subjective health: a benchmark of other health statuses

Treatment

- The education of G3 at age 25: ‘university or more’ vs ‘less than university’
- The education of G1: ‘junior high school or high school’ vs ‘more than high school’

Variables: confounder

Sociodemographic attributes

- Sex, birth cohort and age

Education

- The education of G2: measured by the same way of that of G3

Life course characteristics

- JSEI at the first job, employment status at the first job, unemployment, divorce/widow

Health status before G3 at age 25

- Job leave by health

Data Analytic Plan: Estimand

$$\tau_{x^{\prime}, x} (t) \equiv \mathbf{E}_{\mathcal{S}} \left(\bar{y}_{\mathcal{S}, x^{\prime}}(t) - \bar{y}_{\mathcal{S}, x}(t) \right) \label{eq: estimand}$$

- $\bar{y}$: subjective health of G2
- $x^{\prime}$, x: the level of G1’s education
- t: the level of G3’s education
- $\mathcal{S}$: hypothetical samples

Data Analytic Plan: Identification

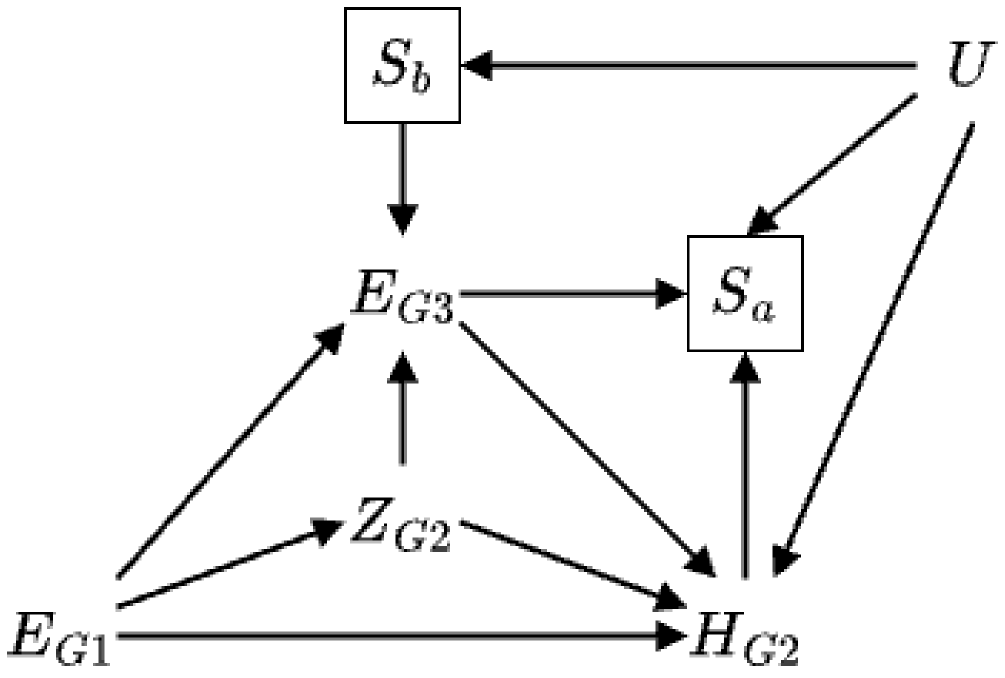

Figure 2: Directed acyclic graph

H_{G2}	subjective health of G3	E_{G1}	education of G1
E_{G3}	education of G3	Z_{G2}	confounders
S_{a}	selection into mortality/hospitalization	S_{b}	selection into fertility

Data Analytic Plan: Estimation

Doubly robust estimation

- a method to estimate the effect of treatment by weighting the treatment probability
- to overcome the issue that the probability of treatment is not assigned randomly
- random forest + cross-validation \[\frac{1}{\Sigma_{i: X_i = x} \hat{w}_i}\Sigma_{i: X_i = x} \hat{w}_i \left(\hat{g}\left(t_i, x_i, \vec{z}_i \right) - y_i \right) \label{eq: doublyrobust} \]

Results

Mean of Subjective Health by G1's Education and G3's Education

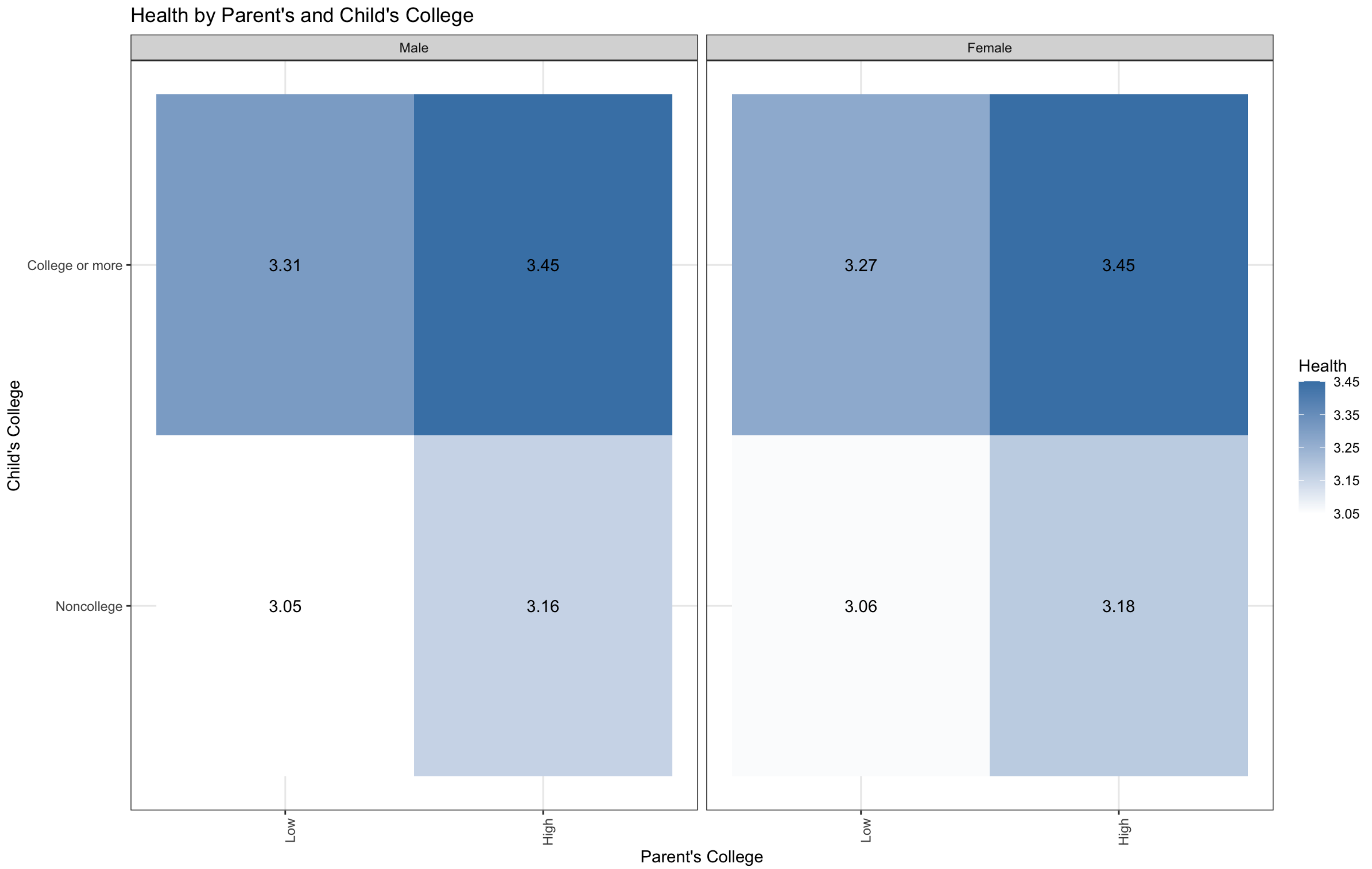

Figure 3: Mean of Subjective Health by G1's Education and G3's Education

The Gap-closing Estimator in All Sample

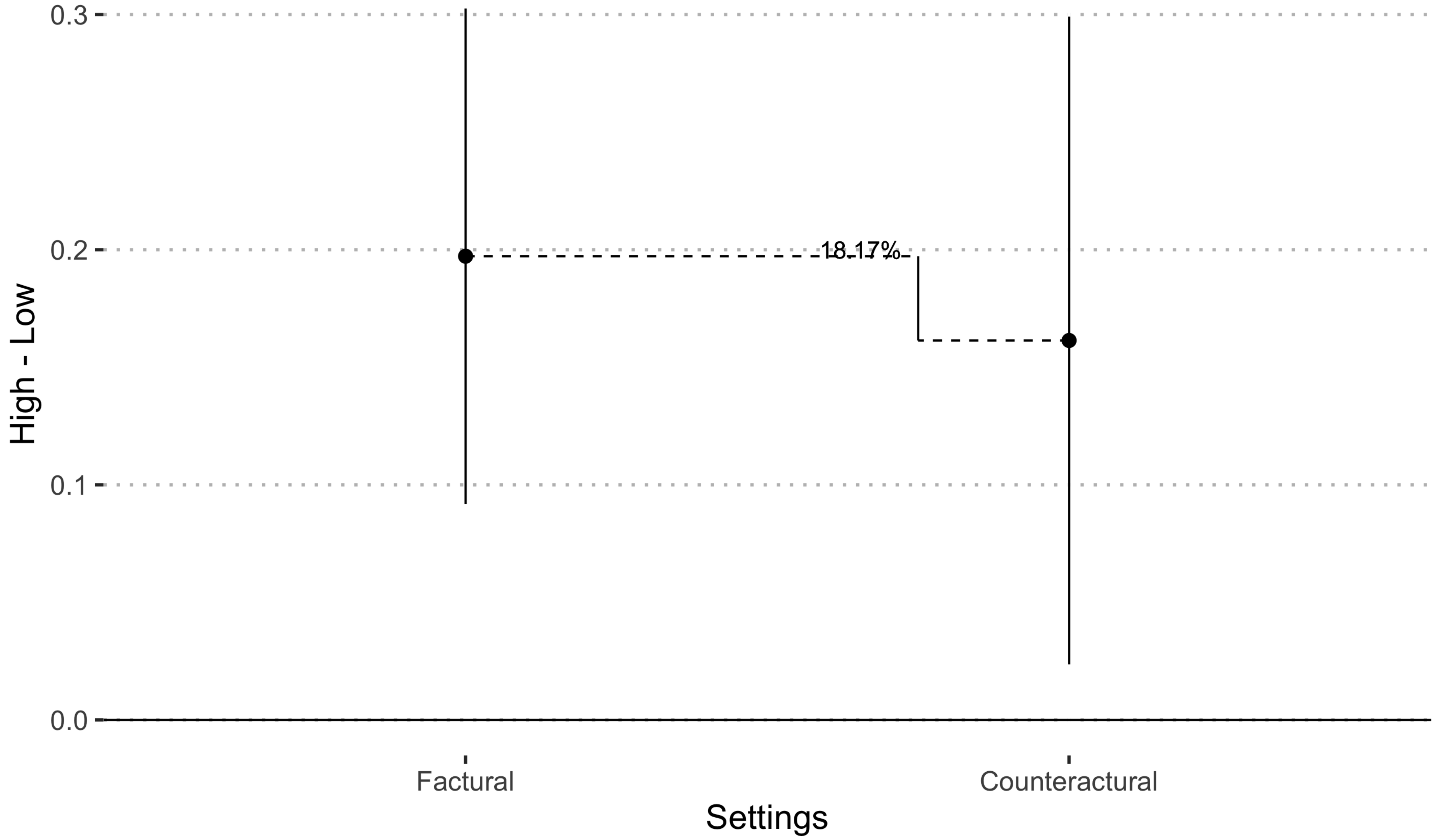

Figure 4: The Gap-closing Estimator in All Sample

The Gap-closing Estimator in Sex

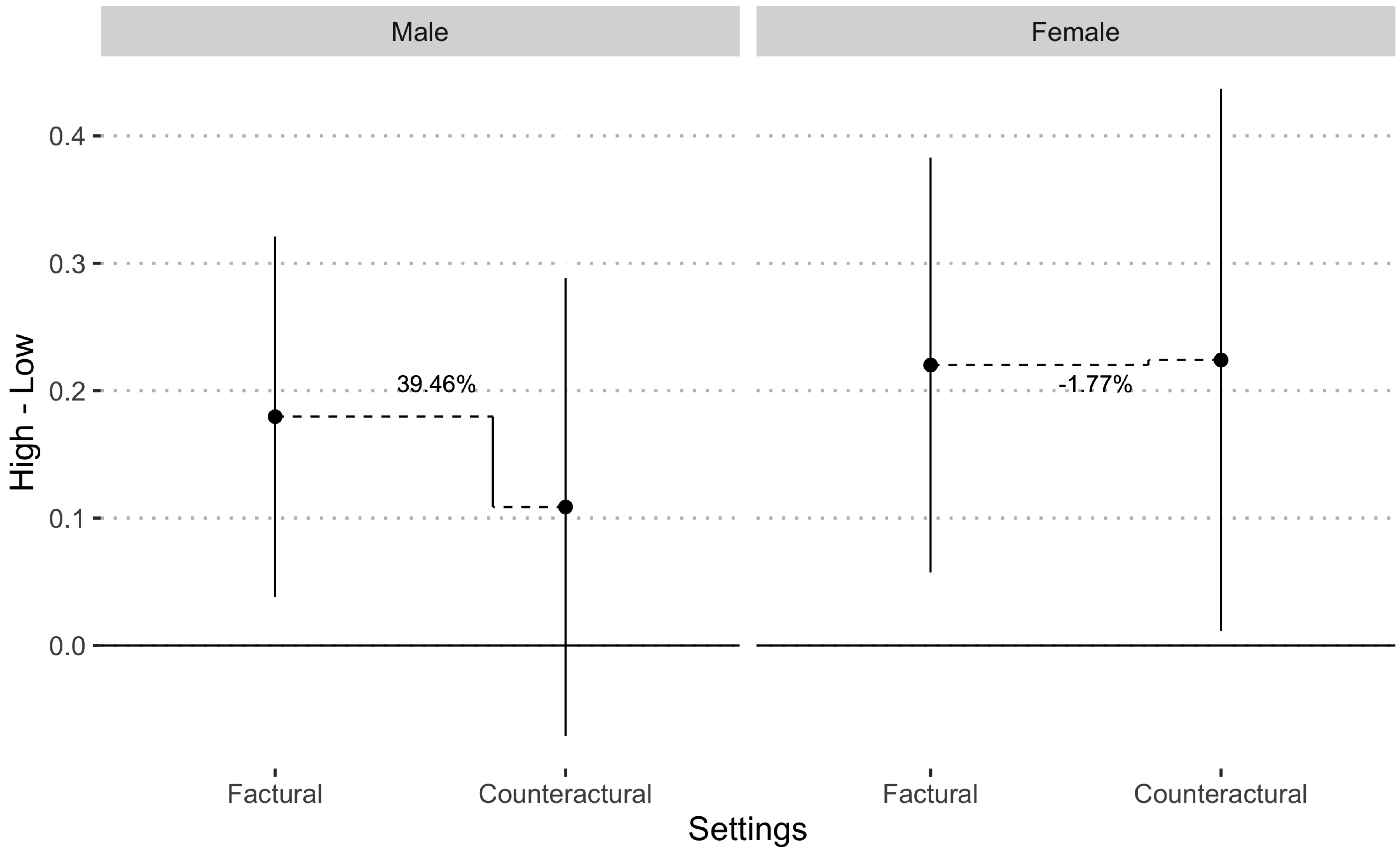

Figure 5: The Gap-closing Estimator by Sex

Discussion

Summary and Discussion

1. 18% of health disparity is closed under the intervention of children's education

- A worthwhile goal: a lower bound for the prospective cohort who shares their lives longer

2. The reduction in disparities is substantive for men(39.5%), but almost none for women(-1.8%)

- Women have broader social networks and obtain the knowledge from others

Health disparity by social origin is modifiable by intervening with the G3's education

- Closing educational inequality for young people and health inequality for older people is not a mutually exclusive agenda

Limitation

1. This study cannot speak to the 'gap-closing' estimand of **specific symptoms** (e.g., cognitive functioning and physical functioning)

- Collecting all health measurements and conducting the outcome-wide longitudinal causal inference (VanderWeele, Mathur, and Chen 2020) can overcome the limitation

2. The sample in SSM2015 is, as in many social surveys, **conditioned by health**

- The gap-closing estimator in this study do not include the serious health problems caused by social origin

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Appendix

Social Causation and Health Selection

The reciprocal relationship between SES and health (e.g., Mulatu and Schooler 2002; Hoffmann, Kröger, and Geyer 2019; Zhang, Xu, and You 2023)

- Health in childhood affects SES, but health in middle and later life is largely determined by health
- health information in childhood or polygenetic index of health to overcome the omitted variable bias

Key assumptions

The health of G2 in childhood affects the education of G3 *only through* the fertility behavior

- The fertility behavior is already conditioned, so I can block the health in childhood and the genetic endowment

Descriptive Statistics

		All	Percent	Mean	SD
	Subjective Health	2530.00	100.00	3.21	0.95
Parental College	Low	2155.00	85.18		
	High	375.00	14.82		
Child College	Non-college	1090.00	43.08		
	College	1440.00	56.92		
Female	Male	1371.00	54.19		
	Female	1159.00	45.81		
	Birth Cohort	2530.00	100.00	1948.77	7.59
	Age	2530.00	100.00	66.23	7.59

Descriptive Statistics(cont.)

		All	Percent	Mean	SD
College	Non-college	2058.00	81.34		
	College	472.00	18.66		
	JSEI at the first job	2530.00	100.00	48.44	7.78
Employment status at the first job	Standard	2138.00	84.51		
	NonStandard	176.00	6.96		
	Self-employed	216.00	8.54		
Unemployment	No	1521.00	60.12		
	Yes	1009.00	39.88		

Descriptive Statistics(cont.)

		All	Percent	Mean	SD
Marital status	No	2054.00	81.19		
	Yes	476.00	18.81		
Job left by health issue	No	2243.00	88.66		
	Yes	287.00	11.34		

Mathematical Representation on Doubly Robust Estimation

Treatment Model

$$ Pr\left(T = t \mid X = x, \vec{Z}_{G2} = \vec{z}, S_a = 1, S_b = 1 \right) $$

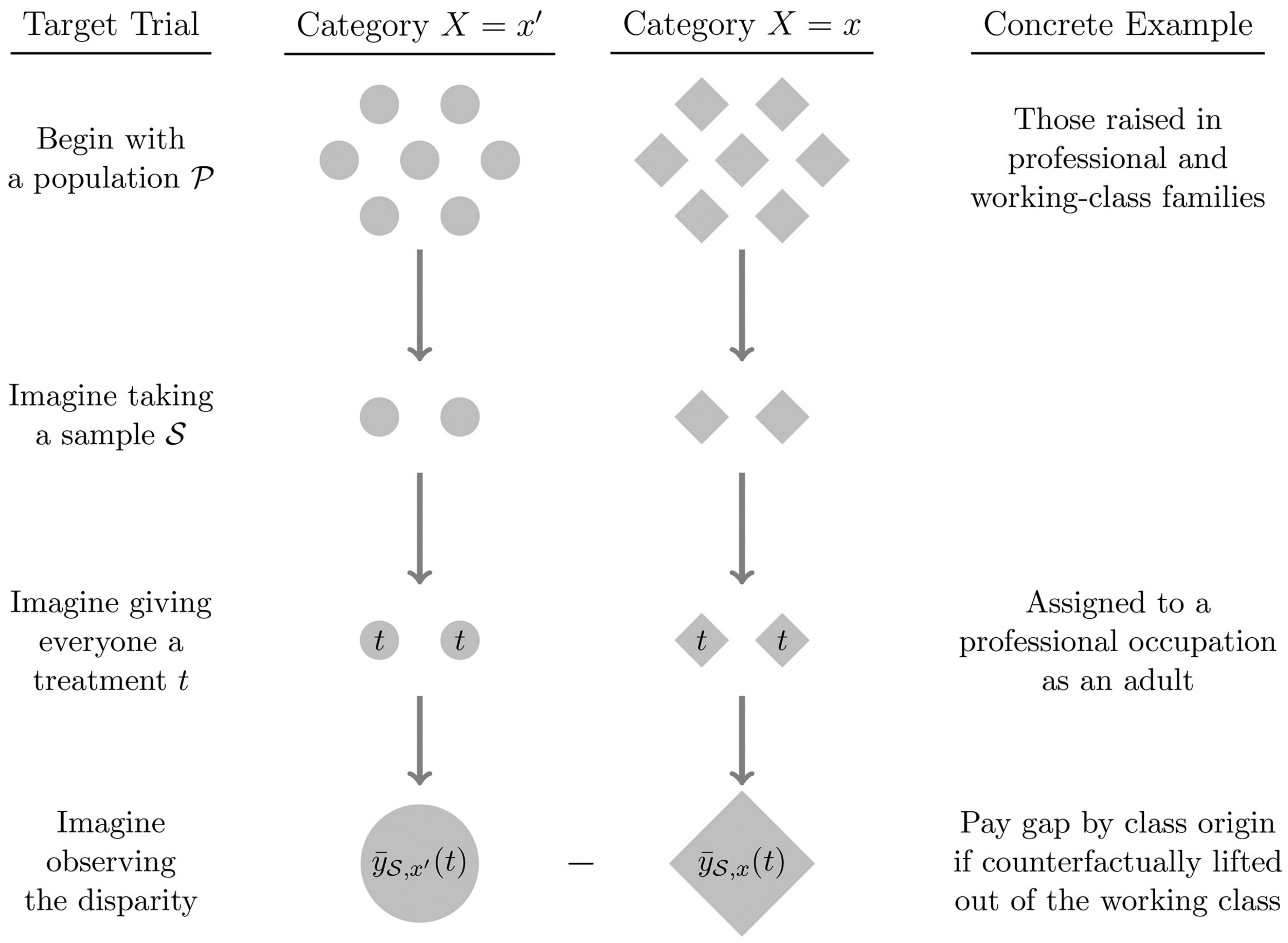
Outcome Model

$$ E\left(Y \mid T = t, X = x, \vec{Z}_{G2} = \vec{z}, S_a = 1, S_b = 1 \right) $$

Doubly Robust Estimation

$$ \frac{1}{\Sigma_{i: X_i = x} \hat{w}_i}\Sigma_{i: X_i = x} \hat{w}_i \left(\hat{g}\left(t_i, x_i, \vec{z}_i \right) - y_i \right) $$

No Counterfactual Intervention on Social Origin



The gap-closing estimand is the expected value of this disparity over hypothetical samples $\mathcal{S}$ from the population $\mathcal{P}$

Realistic Claim

The **scope of the intervention** is a key consideration for gap-closing estimands.

The observational study seeks to approximate a hypothetical experiment (Fig 1).
In that experiment, for whom would we hypothetically reassign the treatment?

Local Extreme
Reassign treatment
for one unit
from each category

Continuum
Reassign treatment for a
sample $\mathcal{S}$ that is some
fraction of the population $\mathcal{P}$

Global Extreme
Reassign treatment
for the entire
population

Tractable Claim
Very close to the
world as observed

Example
Send units i and j
to college

The benefit of college
does not change

Policy-Relevant Claim
In practice, policies apply
to only a small fraction
(at least initially)

Example
Slightly expand the number
of seats available in colleges

The benefit of college
is mostly the same

Speculative Claim
Requires strong theory
about interference

Example
College for all

The benefit of college
might change a lot

Figure 7: Little Interference by Intervening Hypothetical Sample